

WASP-163b

R-2150

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_250520 · created 2026-08-12T01:03:33+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460815.9187 +/- 0.0098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.012
Transit depth (Rp/Rs)^2	2.13 +/- 0.34 [%]
Orbital Inclination (inc)	82.41 +/- 2.74
Airmass coefficient 1 (a1)	0.977 +/- 0.012
Airmass coefficient 2 (a2)	0.015 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	#2 - [31, 419]
Optimal Aperture	3.29
Optimal Annulus	7.22
Transit Duration (day)	0.074 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.146 $\pm$ 0.012 vs 0.119 (deviation 2.25 σ)
Duration fit vs published	0.074 d vs 0.0942 d (deviation 0.78 σ)
Mid-time O–C	24.3 min
Depth SNR	12.2
Residual scatter	1.21 %

Light curve

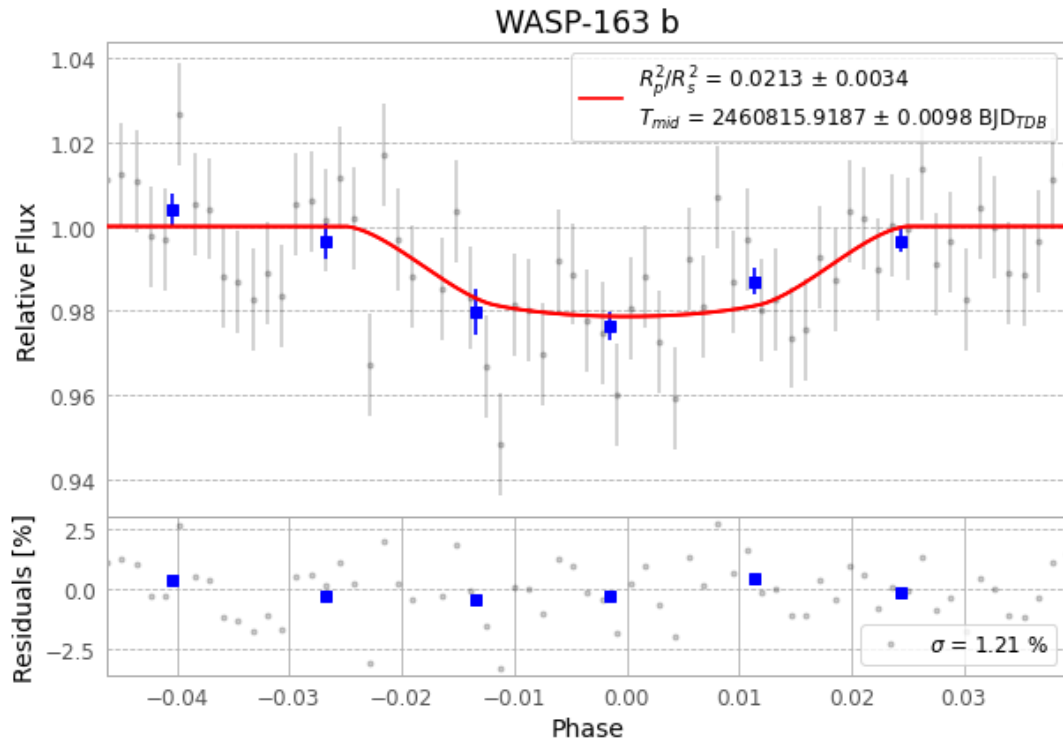


Figure 1 — FinalLightCurve_WASP-163 b_2025-05-20.png (SHA-256 7f76b1d71e5acde3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f09467af2119e1a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), WASP-163250520080615.FITS → WASP-163250520112415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-250520.log (9b641f512332...), FinalLightCurve_WASP-163 b_2025-05-20.png (7f76b1d71e5a...), FinalLightCurve_WASP-163 b_2025-05-20.csv (78b23e31ee5c...)

Compute resources

Wall time 136.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 01:03Z.